

ECON 712: Applied Econometrics

Lecture 2: Regression Fundamentals: Introduction

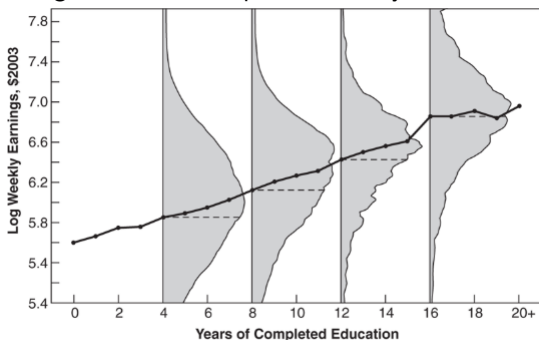
Conditional Expectation Function Decomposition

Multiple Linear Regression

Appendix

Conditional Expectation Function (CEF)

- We can generalize the idea of studying relationships between variables through conditional expectations beyond the binary case.



Raw data and the CEF of average log weekly wages given schooling.

The sample includes white men aged 40–49 in the 1980 IPUMS 5 percent file.

- And, to multiple variables:

$$E[Y_i | \mathbf{X}_i = \mathbf{x}], \quad \mathbf{X}_i = \begin{bmatrix} X_{1i} \\ \vdots \\ X_{ki} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$$

Conditional Expectation Function (CEF)

- The advantage of the CEF is that it decomposes a random variable into two parts:

$$Y_i = E[Y_i | \mathbf{X}_i] + \epsilon_i$$

- **Mean Independence:**

$$E[\epsilon_i | \mathbf{X}_i = \mathbf{x}] = 0$$

- ▶ And, by the LIE:

$$E[\epsilon_i] = E[E[\epsilon_i | \mathbf{X}_i]] = \int E[\epsilon_i | \mathbf{X}_i = \mathbf{x}]g(\mathbf{x}) d\mathbf{x} = 0$$

Notation

- **Uncorrelated:** $E[h(\mathbf{X}_i) \epsilon_i] = 0$

Conditional Expectation Function (CEF)

The CEF decomposition property results in the following two conditions:

1. Among all functions of the random vector $\mathbf{X}_i$, $m(\mathbf{X}_i)$, the one that best predicts Y_i (in the minimum mean squared deviation sense) is:

$$m(\mathbf{X}_i) = E[Y_i | \mathbf{X}_i]$$

- Intuition: $E[Y_i]$ is the best predictor of Y_i , and by the LIE,

$$E[E[Y_i | \mathbf{X}_i]] = E[Y_i].$$

2. ANOVA Theorem:

$$V(Y_i) = V(E[Y_i | \mathbf{X}_i]) + E[V(Y_i | \mathbf{X}_i)].$$

- Variation in Y_i can be decomposed into a part explained by $\mathbf{X}_i$, through conditional expectation, and a part that is not **ANOVA**.

Conditional Expectation Function Decomposition

Multiple Linear Regression

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Multiple Regression: Matrix Form

- We are going to write the Multiple Regression Equation with k regressors and n observations as:

$$\underset{n \times 1}{\mathbf{y}} = \underset{(k+1) \times 1}{\mathbf{X}} \underset{n \times 1}{\boldsymbol{\beta}} + \underset{n \times 1}{\mathbf{u}}$$

where

$$\underset{n \times (k+1)}{\mathbf{X}} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

- By Matrix Multiplication:

$$y_i = \mathbf{X}_i \boldsymbol{\beta} + u_i \text{ (ETM Notation)}$$

$$y_i = \sum_{s=1}^{k+1} x_{is} \beta_s + u_i$$

Multiple Regression

- To fully specify the previous model, we need to put structure on the distribution of the vector of error terms.
 - ▶ ETM defines as a fully specified model, a model that allows us to simulate random vector $\mathbf{Y}$ given $\mathbf{X}$
- Let's assume that u 's are *iid* and that $E[u_i | \mathbf{X}_i = \mathbf{x}] = E[u_i] = 0$ for all possible values of $\mathbf{X}_i$ (the model is still underspecified), then

$$E[Y_i | \mathbf{X}_i = \mathbf{x}] = \mathbf{x}\beta$$

- ▶ Conditional Mean Independence of the error term makes the linear regression to be equal to the CEF
- ▶ Conditional Mean Independence of the error term for all possible values of $\mathbf{X}_i$ can only be achieved if the CEF is linear
- ▶ Note that we have not said anything about whether β is causal (as opposed to what we said about ρ)

Multiple Regression: Matrix Form

- The Method-of-Moment Estimator is expressed as:

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X} \hat{\beta}) = \mathbf{0}$$

- By matrix multiplication:

$$\sum_{i=1}^n x_{ik} (y_i - \mathbf{x}_i \hat{\beta}) = 0 \quad k \text{ equations}$$

$$\sum_{i=1}^n (y_i - \mathbf{x}_i \hat{\beta}) = 0 \quad 1 \text{ equation}$$

- Solving for these, we obtain:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- ▶ $(\mathbf{X}' \mathbf{X})$ can only be inverted if it is full rank (which requires $\mathbf{X}$ to be full rank– i.e. no perfect multicollinearity)

- The same estimator can be derived by solving:

$$\underset{\hat{\beta}}{\text{Min}} (\mathbf{y} - \mathbf{X} \hat{\beta})^T (\mathbf{y} - \mathbf{X} \hat{\beta})$$

Multiple Regression: Matrix Form

- An interesting way to understand regression analysis is to study it using vector spaces
- There are three concepts in vector geometry that are used to study OLS:
 1. Scalar product
 2. Vector Addition
 3. Subspaces of Euclidean Spaces

Scalar Product

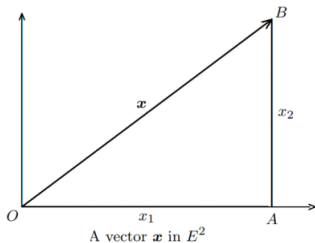
- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

where both vectors are of size $n \times 1$

- An important property application of the scalar product is the **length** or **norm** of a vector defined as:

$$\|\mathbf{x}\| = \left[\sum_{i=1}^n x_i^2 \right]^{\frac{1}{2}} = \langle \mathbf{x}, \mathbf{x} \rangle^{\frac{1}{2}}$$



Scalar Product

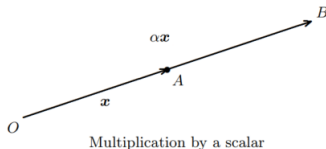
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- Another important property is that if the angle between two vectors, θ , is zero, then:
 - ▶ This property is based on multiplying a vector by a scalar:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\|$$



Scalar Product

- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

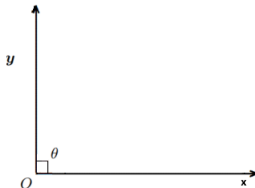
$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

where both vectors are of size $n \times 1$

- Lastly if the angle between two vectors, $\theta = 90^\circ$ (i.e. they are **orthogonal**), then:

► E.g. $\mathbf{x}^T = [1, 0]$, $\mathbf{y}^T = [0, 1]$

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0$$



Conditional Expectation Function Decomposition

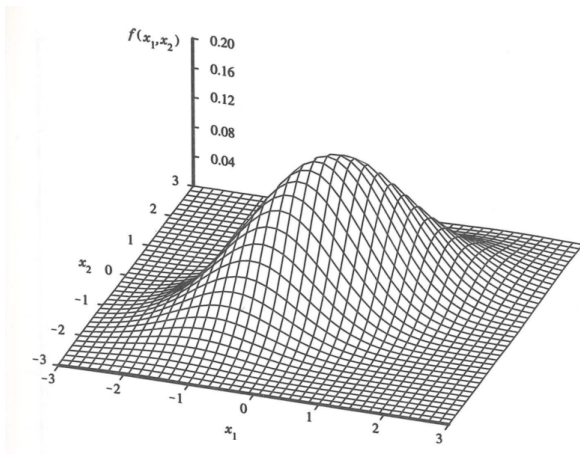
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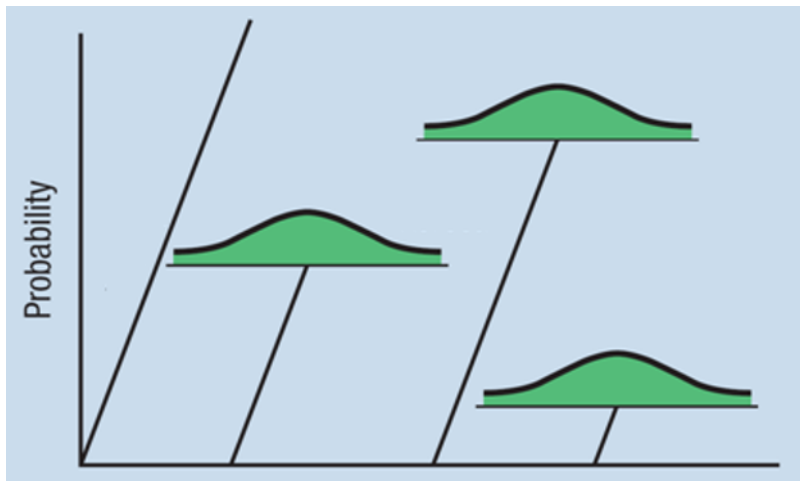
Short-Hand Notation

- Example with two random variables: X_1 , X_2 :

$$E[\epsilon_i] = \int \int E[\epsilon_i \mid X_{1i} = x_1, X_{2i} = x_2] g(x_1, x_2) dx_1 dx_2$$



ANOVA Representation



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